Prob. 1 – Polynomial interpolation Ex2 – K.P.Lind

$$f(x) = 3^{-(x-1)^2}$$

a) Interpolate $f(x)$ by a polynomial of minimal degree with the given points

Using Lagrange interpolation method: $\boxed{l_j(x) = \prod_{\substack{m=0\\m\neq j}}^{k} \frac{x - x_m}{x_j - x_m}}$ $\boxed{L(x) = \sum_{j=0}^{k} l_j\, y_j}$

| | | | | |
|---|---|---|---|---|
| | $y_0$ | $y_1$ | $y_2$ | $y_3$ |
| $y$ | $1/81$ | $1/3$ | $1$ | $1/3$ |
| $x$ | $-1$ | $0$ | $1$ | $2$ |
| | $x_0$ | $x_1$ | $x_2$ | $x_3$ |

$$l_0 = \prod_{\substack{m=0\\m\neq 0}}^{k=3} \frac{x-x_m}{x_0-x_m} = \frac{x-x_1}{x_0-x_1}\cdot\frac{x-x_2}{x_0-x_2}\cdot\frac{x-x_3}{x_0-x_3} = \frac{x-0}{-1-0}\cdot\frac{x-1}{-1-1}\cdot\frac{x-2}{-1-2} = \frac{x^3}{6} - \frac{x^2}{2} + \frac{x}{3}$$

$$l_1 = \prod_{\substack{m=0\\m\neq 1}}^{k=3} \frac{x-x_m}{x_1-x_m} = \frac{x-x_0}{x_1-x_0}\cdot\frac{x-x_2}{x_1-x_2}\cdot\frac{x-x_3}{x_1-x_3} = \frac{x-(-1)}{0-(-1)}\cdot\frac{x-1}{0-1}\cdot\frac{x-2}{0-2} = \frac{x^3}{2} - x^2 - \frac{x}{2} + 1$$

$$l_2 = \prod_{\substack{m=0\\m\neq 2}}^{k=3} \frac{x-x_m}{x_2-x_m} = \frac{x-x_0}{x_2-x_0}\cdot\frac{x-x_1}{x_2-x_1}\cdot\frac{x-x_3}{x_2-x_3} = \frac{x-(-1)}{1-(-1)}\cdot\frac{x-0}{1-0}\cdot\frac{x-2}{-2} = -\frac{x^3}{4} + \frac{x^2}{4} + \frac{x}{2}$$

$x^2 - 2x -$

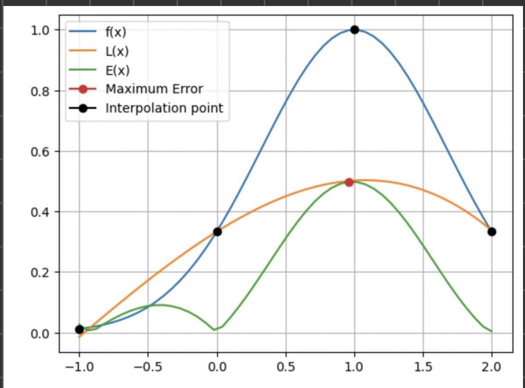
$$l_3 = \prod_{\substack{m=0\\m\neq 3}}^{k=3} \frac{x-x_m}{x_3-x_m} = \frac{x-x_0}{x_3-x_0}\cdot\frac{x-x_1}{x_3-x_1}\cdot\frac{x-x_2}{x_3-x_2} = \frac{x-(-1)}{2-(-1)}\cdot\frac{x-0}{2-0}\cdot\frac{x-1}{2-1} = \frac{x^3}{6} - \frac{x}{6}$$

$$L(x) = \frac{1}{81}\left(\frac{x^3}{6} - \frac{x^2}{2} + \frac{x}{3}\right) + \frac{1}{3}\left(\frac{x^3}{2} - x^2 - \frac{x}{2} + 1\right) + 1\left(-\frac{x^3}{4} + \frac{x^2}{4} + \frac{x}{2}\right) + \frac{1}{3}\left(\frac{x^3}{6} - \frac{x}{6}\right)$$

$$\underline{L(x) = -\frac{25}{972}x^3 - \frac{29}{324}x^2 + \frac{23}{81}x + \frac{1}{3}}$$

$$\|E(x)\|_\infty = \|f(x) - p_n(x)\|_\infty$$

$x \in [-1,2]$:

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We can find the maximum error in the point x =  0.9592
The value is E(x) =  0.4975
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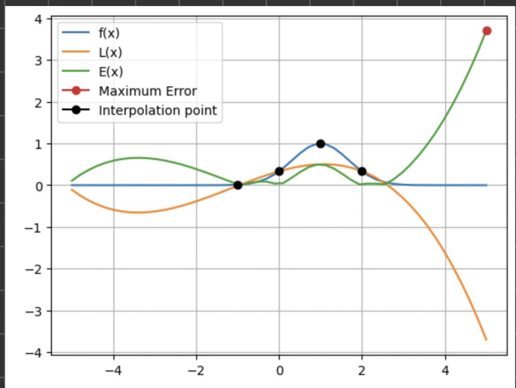
$x \in [-5,5]$:

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We can find the maximum error in the point x =  5.0
The value is E(x) =  3.6996
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We can see that the error grows a lot in line with the interval

b) Find the Gaussian nodes on the interval $[-1,2]$ for $n=3$

$$\boxed{x_i = \frac{1}{2}(a+b) + \frac{1}{2}(b-a)\tilde{x}_i}$$

$$x_0 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\,\tilde{x}_0 = 1,6699$$

$$x_1 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\,\tilde{x}_1 = 1,3300$$

$$x_2 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\,\tilde{x}_2 = 1{,}9306$$

$$x_3 = \frac{1}{2}(-1+2) + \frac{1}{2}(2-(-1))\,\tilde{x}_3 = 1{,}0694$$

c) Interpolate $f(x)$ in the Gaussian nodes for $n=3$. What is max error on the intervals $[-1,2]$ and $[-5,5]$? Plot $f(x)$ and the int. poly.

| | $y_0$ | $y_1$ | $y_2$ | $y_3$ |
|---|---|---|---|---|
| y | 0,61 | 0,89 | 0,39 | 0,99 |
| x | 1,67 | 1,33 | 1,93 | 1,07 |
| | $x_0$ | $x_1$ | $x_2$ | $x_3$ |

$$l_0 = \prod_{\substack{m=0\\m\neq 0}}^{k=3} \frac{x-\tilde{x}_m}{\tilde{x}_0-\tilde{x}_m} = \frac{x-\tilde{x}_1}{\tilde{x}_0-\tilde{x}_1}\cdot\frac{x-\tilde{x}_2}{\tilde{x}_0-\tilde{x}_2}\cdot\frac{x-\tilde{x}_3}{\tilde{x}_0-\tilde{x}_3}$$

$$l_1 = \prod_{\substack{m=0\\m\neq 1}}^{k=3} \frac{x-\tilde{x}_m}{\tilde{x}_1-\tilde{x}_m} = \frac{x-\tilde{x}_0}{\tilde{x}_1-\tilde{x}_0}\cdot\frac{x-\tilde{x}_2}{\tilde{x}_1-\tilde{x}_2}\cdot\frac{x-\tilde{x}_3}{\tilde{x}_1-\tilde{x}_3}$$

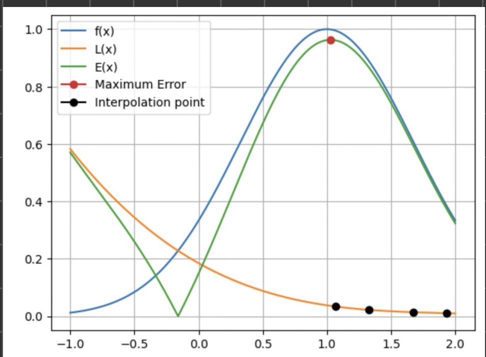
$$l_2 = \prod_{\substack{m=0\\m\neq 2}}^{k=3} \frac{x-\tilde{x}_m}{\tilde{x}_2-\tilde{x}_m} = \frac{x-\tilde{x}_0}{\tilde{x}_2-\tilde{x}_0}\cdot\frac{x-\tilde{x}_1}{\tilde{x}_2-\tilde{x}_1}\cdot\frac{x-\tilde{x}_3}{\tilde{x}_2-\tilde{x}_3}$$

$$l_3 = \prod_{\substack{m=0\\m\neq 3}}^{k=3} \frac{x-\tilde{x}_m}{\tilde{x}_3-\tilde{x}_m} = \frac{x-\tilde{x}_0}{\tilde{x}_3-\tilde{x}_0}\cdot\frac{x-\tilde{x}_1}{\tilde{x}_3-\tilde{x}_1}\cdot\frac{x-\tilde{x}_2}{\tilde{x}_3-\tilde{x}_2}$$

$$\underline{L(x) = f(\tilde{x}_0)l_0 + f(\tilde{x}_1)l_1 + f(\tilde{x}_2)l_2 + f(\tilde{x}_3)l_3}$$

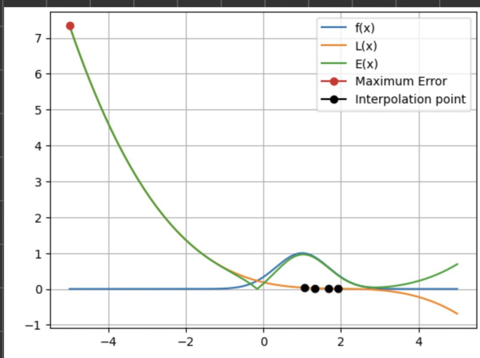
$$\|E(x)\|_\infty = \|f(x) - p_n(x)\|_\infty$$

$x \in [-1,2]$:




We can find the maximum error in the point x = 1.027
The value is E(x) = 0.9629

$x \in [-5,5]$:




We can find the maximum error in the point x = −5.0
The value is E(x) = 7.3387

Prob. 2 – Lagrange interpolation  Ex2 – K.P. Lind

a) Compute the cardinal func. $l_0(x), l_1(x), l_2(x)$ explicity for the points $x_0 = -1$

$x_1 = 1 \quad x_2 = 5$

$$l_0 = \prod_{\substack{m=0\\m\neq 0}}^{k=2} \frac{x - x_m}{x_0 - x_m} = \frac{x - x_1}{x_0 - x_1} \cdot \frac{x - x_2}{x_0 - x_2} = \frac{x-1}{-1-1} \cdot \frac{x-5}{-1-5} = \frac{x^2}{8} - \frac{3x}{4} + \frac{5}{8}$$

$$l_1 = \prod_{\substack{m=0\\m\neq 1}}^{k=2} \frac{x - x_m}{x_1 - x_m} = \frac{x - x_0}{x_1 - x_0} \cdot \frac{x - x_2}{x_1 - x_2} = \frac{x-(-1)}{1-(-1)} \cdot \frac{x-5}{1-5} = -\frac{x^2}{8} + \frac{x}{2} + \frac{5}{8}$$

$$l_2 = \prod_{\substack{m=0\\m\neq 2}}^{k=2} \frac{x - x_m}{x_2 - x_m} = \frac{x - x_0}{x_2 - x_0} \cdot \frac{x - x_1}{x_2 - x_1} = \frac{x-(-1)}{5-(-1)} \cdot \frac{x-1}{5-1} = \frac{x^2}{24} - \frac{1}{24}$$

b) Verify that the cardinal functions satisfy the equations

$l_i(x_i) = 1$ for all $i = 0, \dots, k$

$l_i(x_j) = 0$ for all $i, j = 0, \dots, k$ with $i \neq j$

If we replace $x = x_n$ when we calculate the cardinal functions $l_i(x_i)$ we get

$$l_i(x_i) = \prod_{\substack{m=0\\m\neq n}}^{k} \frac{x - x_m}{x_n - x_m} \Rightarrow l_i(x_i) = \prod_{\substack{m=0\\m\neq n}}^{k} \frac{x_n - x_m}{x_n - x_m} = 1$$

we can do kind of the same thing when proving the other equation.

If we replace $x = x_m$ when we calculate the cardinal function $l_i(x_j)$ we get

$$l_i(x_j) = \prod_{\substack{m=0\\m\neq n}}^{k} \frac{x - x_m}{x_n - x_m} \Rightarrow l_i(x_m) = \prod_{\substack{m=0\\m\neq n}}^{k} \frac{x_m - x_m}{x_n - x_m} = 0$$

c) Assume we are given points $(x_i, y_i)_{i=0}^{n}$ with distinct values $x_i$. Show that the interpolation polynomial $p_n$ given by

$p_n(x) = \sum_{i=0}^{n} y_i \, l_i(x)$

If we take a look at the function for $l_i(x_i)$, we can notice that

$l_i$ is the prod. of $n$ linear poly. $\Rightarrow$ $l_i$ is a poly. of degree $n$.

Additionaly, if we take a look at our proof above, we can see that

$l_i(x_i) = 1$, which leads us to

$p_n(x_i) = \sum_{i=0}^{n} y_i \, l_i(x) = y_i$

Prob. 3 – Trapezoidal rule 

$f(x) = \sqrt{x}$ , $x \in [0, 2]$

a) Implement the composite trapezoidal rule for the numerical computation of this integral

Using the composite trapezoidal rule we get:

$$\int_a^b f(x)\,dx = \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right)$$

We choose $n = 4 \Rightarrow h = \frac{b-a}{n} = 0{,}5 \Rightarrow x_i = \{\overset{x_0}{0}, \overset{x_1}{0.5}, \overset{x_2}{1}, \overset{x_3}{1.5}, \overset{x_4}{2}\}$

$$\int_0^2 \sqrt{x}\,dx = \frac{0.5}{2}\left(f(0) + 2f(0.5) + 2f(1) + 2f(1.5) + f(2)\right)$$

$$= \frac{1}{4}\left(\sqrt{0} + 2\sqrt{0.5} + 2\sqrt{1} + 2\sqrt{1.5} + \sqrt{2}\right)$$

$$= \underline{\underline{1{,}82}}$$

b) Estimate the numerical converge rate for this method. Assume that the error satisfies

$$err_m := \left| T_m(0,1) - \int_0^1 \sqrt{x}\,dx \right| \approx Ch^p,$$

and estimate numerically the number p.

We calculate the exact value of the integral to be $\int_0^1 \sqrt{x}\,dx = \frac{2}{3}$

If we use what we know about numerical verification

$$p \approx \frac{\log(e(h_{m+1})/e(h_m))}{\log(h_{m+1}/h_m)},$$

and we choose a rather high value for m $\overset{300}{''}$ , $p \approx 1{,}5$.

c) We know that the value of p should be $p = 2$ if the function f is two times continuosly differensiable. This will not hold for our $f'(0)$ (not continuous).

Prob. 4 – Runge phenomenon  Ex2 – K.P. Lind

$$f(x) = \frac{1}{1+25x^2}$$

a) Interpolate $f(x)$ by a polynomial of minimal degree, using the given points

Using Lagrange interpolation method: $\boxed{l_j(x) = \prod_{\substack{m=0 \\ m\neq j}}^{n} \frac{x-x_m}{x_j-x_m}}$ $\boxed{L(x) = \sum_{j=0}^{n} l_j y_i}$

| | $y_0$ | $y_1$ | $y_2$ |
|---|---|---|---|
| y | 1/26 | 1 | 1/26 |
| x | -1 | 0 | 1 |
| | $x_0$ | $x_1$ | $x_2$ |

$$l_0 = \prod_{\substack{m=0 \\ m\neq 0}}^{k=2} \frac{x-x_m}{x_0-x_m} = \frac{x-x_1}{x_0-x_1} \cdot \frac{x-x_2}{x_0-x_2} = \frac{x-0}{-1-0} \cdot \frac{x-1}{-1-1} = \frac{x^2}{2} - \frac{x}{2}$$

$$l_1 = \prod_{\substack{m=0 \\ m\neq 1}}^{k=2} \frac{x-x_m}{x_1-x_m} = \frac{x-x_0}{x_1-x_0} \cdot \frac{x-x_2}{x_1-x_2} = \frac{x-(-1)}{0-(-1)} \cdot \frac{x-1}{0-1} = 1-x^2$$

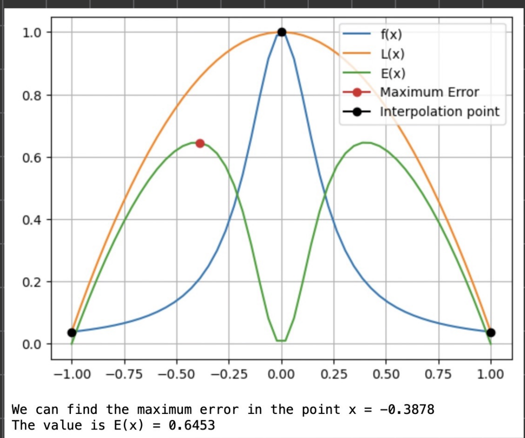
$$l_2 = \prod_{\substack{m=0 \\ m\neq 2}}^{k=2} \frac{x-x_m}{x_2-x_m} = \frac{x-x_0}{x_2-x_0} \cdot \frac{x-x_1}{x_2-x_1} = \frac{x-(-1)}{1-(-1)} \cdot \frac{x-0}{1-0} = \frac{x^2}{2} + \frac{x}{2}$$

$$L(x) = (1/26)\left(\frac{x^2}{2} - \frac{x}{2}\right) + (1-x^2) + (1/26)\left(\frac{x^2}{2} + \frac{x}{2}\right)$$

$$\underline{L(x) = 1 - 25/26\, x^2}$$

$$\| E(x) \|_\infty = \| f(x) - p_n(x) \|_\infty$$

$x \in [-1, 1]$:




b) The Chebyshev nodes on an interval $[a,b]$ are def. as

$$\tilde{x}_i = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{(2i+1)\pi}{2(n+1)}\right), \quad i = 0, 1, 2, \dots, n.$$

Interpolate $f(x)$ in the Chebyshev nodes for $n=2$. What is the maximal error on the int $[-1,1]$? Plot $f(x)$ and the inter. poly.

$$\tilde{x}_0 = \frac{(-1)+1}{2} + \frac{1-(-1)}{2} \cos\left(\frac{(2\cdot 0+1)\pi}{2(2+1)}\right) = \cos(\pi/6) = \sqrt{3}/2 = 0{,}866$$

$$\tilde{x}_1 = \frac{(-1)+1}{2} + \frac{1-(-1)}{2} \cos\left(\frac{(2\cdot 1+1)\pi}{2(2+1)}\right) = \cos(\pi/2) = 0$$

$$\tilde{x}_2 = \frac{(-1)+1}{2} + \frac{1-(-1)}{2} \cos\left(\frac{(2\cdot 2+1)\pi}{2(2+1)}\right) = \cos(5\pi/6) = -\sqrt{3}/2 = -0{,}866$$

| | $y_0$ | $y_1$ | $y_2$ |
|---|---|---|---|
| $y$ | $4/75$ | $1$ | $4/75$ |
| $x$ | $\sqrt{3}/2$ | $0$ | $-\sqrt{3}/2$ |
| | $x_0$ | $x_1$ | $x_2$ |

$$L_0 = \prod_{\substack{m=0\\m\neq 0}}^{k=2} \frac{x-x_m}{x_0-x_m} = \frac{x-x_1}{x_0-x_1}\cdot\frac{x-x_2}{x_0-x_2} = \frac{x-0}{\sqrt{3}/2-0}\cdot\frac{x-(-\sqrt{3}/2)}{\sqrt{3}/2-(-\sqrt{3}/2)} = \frac{2x^2}{3}+\frac{x}{\sqrt{3}}$$

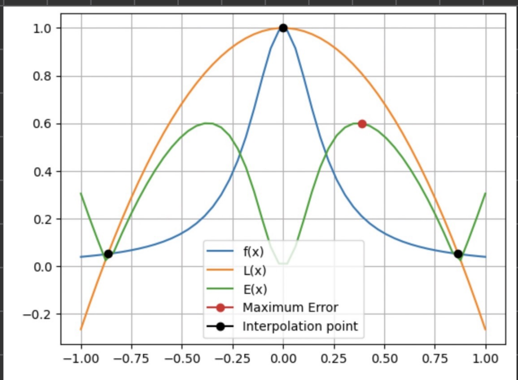
$$L_1 = \prod_{\substack{m=0\\m\neq 1}}^{k=2} \frac{x-x_m}{x_1-x_m} = \frac{x-x_0}{x_1-x_0}\cdot\frac{x-x_2}{x_1-x_2} = \frac{x-\sqrt{3}/2}{0-\sqrt{3}/2}\cdot\frac{x-(-\sqrt{3}/2)}{0-(-\sqrt{3}/2)} = \frac{-4x^2}{3}-1$$

$$L_2 = \prod_{\substack{m=0\\m\neq 2}}^{k=2} \frac{x-x_m}{x_2-x_m} = \frac{x-x_0}{x_2-x_0}\cdot\frac{x-x_1}{x_2-x_1} = \frac{x-\sqrt{3}/2}{(-\sqrt{3}/2)-\sqrt{3}/2}\cdot\frac{x-0}{(-\sqrt{3}/2)-0} = -\frac{2x^2}{3}+\frac{x}{\sqrt{3}}$$

$$h(x) = (4/75)\left(\frac{2x^2}{3}+\frac{x}{\sqrt{3}}\right) + \left(-\frac{4x^2}{3}-1\right) + (4/75)\left(-\frac{2x^2}{3}+\frac{x}{\sqrt{3}}\right)$$

$$\|E(x)\|_\infty = \|f(x)-p_n(x)\|_\infty$$

$x \in [-1, 1]$:

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We can find the maximum error in the point x = 0.3878
The value is E(x) = 0.5995
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Prob. 5

I dont even know what a knot vector is